

Technical Report: Sales Analytics | by Okeoghene | Medium

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Technical Report: Sales Analytics Introduction At my current internship, I was tasked with conducting an initial insight analysis on a retail dataset. The dataset was mostly clean with minimal null values or empty cells, especially in key performance points. However, there were inconsistencies in datatype distribution and some pre-existing calculations (Sales column). Although this task was intended to be a first-glance analysis rather than an in-depth study, basic data cleaning was necessary to ensure accurate results. This included creating a calendar table and correcting data types, particularly for date-time fields.

Observations The Sales Analytics Dashboard provides a comprehensive overview of sales performance across multiple dimensions. The following key metrics were identified:

- 1. Key Metrics:** Number of Customers: 92 Total Number of Orders: 2,823 Total Revenue: 8.29 million
- 2. Sales Trend Analysis:** The sales trend over the year, visualized as a line graph, reveals a significant spike in November, likely due to seasonal promotions or holidays. This insight is crucial for planning future sales campaigns and inventory management.
- 3. Top Customers by Revenue:** The highest revenue was generated by "Collectable Mini Designs Co." (58,018.57), followed by "Dragon Souvenirs, Ltd." (57,334.56) and "Technics Stores Inc." (55,176.87). Identifying these key accounts helps in focusing customer retention strategies and understanding their purchasing behaviors.
- 4. Top Salespersons by Revenue:** Freyre leads with 766,195.05 in revenue, followed by Nelson (597,751.56) and Young (348,077.64). Recognizing these top performers assists in developing reward programs and replicating successful sales techniques across the team.
- 5. Revenue Distribution by City:** Madrid is the leading city in revenue generation, followed by San Rafael and NYC. This geographic analysis aids in targeting regional marketing efforts and efficient resource allocation.
- 6. Order Status Distribution:** The status of orders is predominantly shipped (92.28%), indicating efficient order fulfillment processes. A small percentage of orders are cancelled (2.06%), on hold, or in process, highlighting areas that could benefit from operational improvements.
- 7. Revenue by Product Line:**— Classic Cars lead in revenue contribution at 35.8%, followed by Vintage Cars and Motorcycles. This breakdown assists in inventory planning and marketing focus, ensuring high-demand products are adequately stocked and promoted.

Conclusion The Sales Analytics Dashboard effectively summarizes sales data, enabling strategic decisions to optimize performance, enhance customer relationships, and streamline operations. By addressing initial data inconsistencies and performing essential data cleaning, the insights derived were both accurate and actionable. This comprehensive analysis highlights current performance metrics and provides a foundation for future sales strategies and operational improvements. It is important to note also that while this was first-look analysis, there were difficulties understanding a few columns, including the Sales and MSRP. With a lack of data dictionary, it was difficult to work with sole figures in these columns.